

Program Question_4

```
Implicit none
Real::t,h1,h2,h3,h4,i1,i2,i3,i4

!For input
Open(Unit=31,File='in_a4q1.txt')
Open(Unit=35,File='out_a4q1.txt')
Read(31,*)t,h1,h2,h3,h4
!Three-point midpoint formula for h1
i1=(q(t+h1)-q(t-h1))/(2.0*h1)
!Three point endpoint formula for h2
i2=(-3*q(t+h2)+4*q(t+h2)-q(t+2*h2))/(2.0*h2)
!Five point mid-point formula for h3
i3=(q(t-2*h3)-8*q(t-h3)+8*q(t+h3)-q(t+2*h3))/(12.0*h3)
!Five point- endpoint formula for h4
i4=(-25*q(t)+48*q(t+h4)-36*q(t+2*h4)+16*q(t+3*h4)-3*q(t+4*h4))/(12.0*h4)
!For output
Write(35,*)"Numerical Differentiation of i(t) at t=5s"
Write(35,*)" Method |Approximation_i(t)"
Write(35,'(A40,F12.6)') "3-point midpoint formula (h=-0.25) |", i1
Write(35,'(A40,F12.6)') "3-point endpoint formula (h=0.25) |", i2
Write(35,'(A40,F12.6)') "5-point midpoint formula (h=.50) |", i3
Write(35,'(A40,F12.6)') "5-point endpoint formula (h=0.50) |", i4

contains
Function q(t) result(q_val)
  Implicit none
  Real,Intent(In)::t
  Real::q_val
  q_val=60.0-60.0*exp(-t/5.0)
End function
```

End Program